

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

SATELLITE COMMUNICATIONS
(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Mention the parameters used in orbital analysis of a satellite. | 2M |
| (b) Why uplink and down link frequencies are different for a satellite system? | 2M |
| (c) Define reliability of a device or subsystem. | 2M |
| (d) Write the various losses to be considered for a satellite link. | 2M |
| (e) Write the advantages of CDMA technique. | 2M |
| (f) Define the efficiency of TDMA and write the expression for it. | 2M |
| (g) Name some mobile satellite system. | 2M |
| (h) Which kinds of communication can be accommodated by the network structure of INTELSAT? | 2M |
| (i) What are the various types of power amplifiers used at satellite earth station? | 2M |
| (j) What is antenna pointing loss? Explain. | 2M |

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) What are the various orbital effects in satellite communication system performance? Explain. 5M
(b) What is azimuth angle with respect to a satellite? Derive the expression for it. 5M
- OR**
- 3 (a) What are the look angles with respect to a satellite? Explain with relevant diagrams. 5M
(b) A satellite is in an elliptical orbit with a perigee of 10200 km and an apogee of 4200 km. using a mean earth radius of 6378.14 km, find the period of the orbit. 5M
- 4 (a) What are the various satellite subsystems? Explain communication subsystem with a neat block diagram. 5M
(b) Explain the redundancy type of approach used for improving equipment reliability in satellite. 5M
- OR**
- 5 (a) Narrate the process of the design of downlink in satellite communications. 5M
(b) Discuss about spacecraft subsystems. 5M
- 6 (a) Show the operation of FDMA and list how this differs from FDM. 5M
(b) Calculate the frame efficiency for an INTELSAT frame given the following information: 5M
Total frame length = 120832 symbols, Traffic bursts per frame = 14, Reference bursts per frame = 2 and Guard interval = 103 symbols.
- OR**
- 7 (a) Point out the comment on pros and cons of satellite system based on TDMA. Also explain the TDMA frame format in detail with relevant diagrams. 5M
(b) Distinguish between TDMA, FDMA and CDMA techniques. 5M

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- 8 (a) Calculate the path loss for a satellite to earth down link with a distance of 38,500 km at a frequency of 4.0 GHz. What is the uplink path loss if the earth station transmits to the same satellite at a frequency of 6.0 GHz? 5M
(b) Write short notes on INMARSAT. 5M
- OR**
- 9 (a) For a given satellite and signal transmission, identify the earth station parameters affecting the C/N ratio. 5M
(b) A satellite is in an elliptical orbit with a perigee of 1000 km and an apogee of 4000 km. find the period of the orbit and eccentricity of the orbit. 5M
- 10 (a) Draw the transmitter and receiver block diagrams of an earth station and explain its Working. 5M
(b) Write short notes on different types of antennas employed for satellite communication. 5M
- OR**
- 11 (a) Distinguish about the Low-noise amplifier and High power amplifier in satellite earth station. 5M
(b) Explain direct broad casting TV. 5M
